higher education & training

Department:
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NATIONAL CERTIFICATE

COMMUNICATION-ELECTRONICS N6

(8080246)

9 April 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages and a formula sheet of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
COMMUNICATION-ELECTRONICS N6
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Sketches must be large, neat and fully labelled.
5. Write neatly and legibly.

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QUESTION 1: FILTERS

1.1 Design a T-section filter network that will eliminate all frequencies between the ranges of 7,5 kHz and 10 kHz for a transmission line of 600 ohms. The drawing of the completed network must show ALL relevant values. (6)

1.2 Draw a neat, labelled graph of attenuation versus frequency for each of the following:

1.2.1 Band-stop filter

1.2.2 Band-pass filter

(2 × 2) (4)

1.3 A high-pass π-section filter with a cut-off frequency of 3 kHz and a frequency of infinite attenuation of 1,8 kHz is used on a transmission line of 600 ohms.

Obtain the terminating input and output half-sections and draw the completed filter. Show ALL the relevant component values. (7) **[17]**

QUESTION 2: ATTENUATORS

2.1 Determine the values of Z_1, Z_2 and Z_3 in the attenuator network in FIGURE 1 if the generator has to supply one-third of the current to the load R_L. (5)

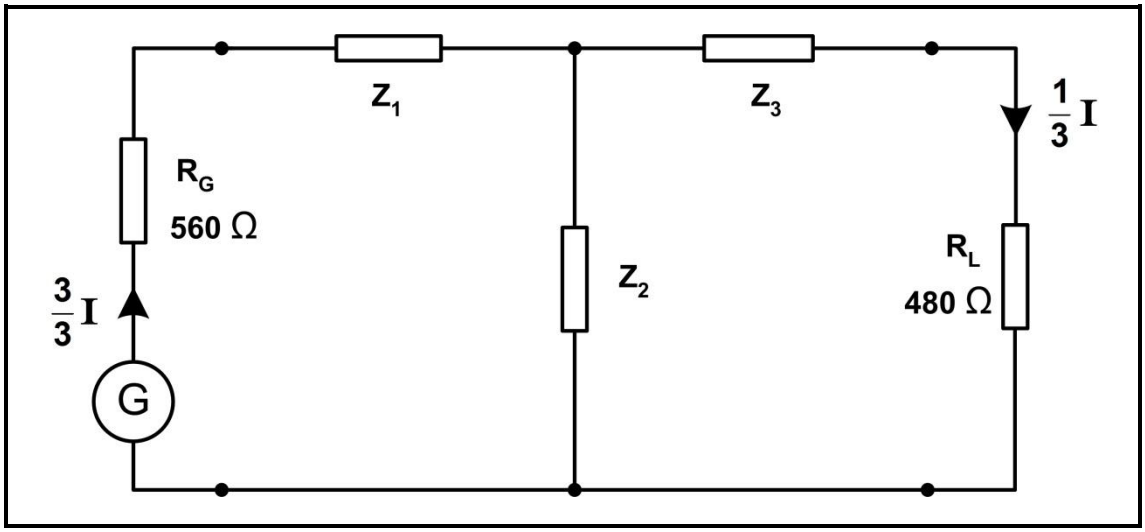


FIGURE 1

2.2 Impedance matching and possible attenuation are important concepts in communication systems. A certain device with an output impedance of 350 ohms has to be matched to a device with an input impedance of 500 ohms.

Design a π-type attenuator of which the attenuation figure may not exceed 15. Draw the complete network and insert ALL the relevant component values. (8)

2.3 Draw the following fully labelled balanced attenuator networks:

2.3.1 π-type

2.3.2 L-type

2.3.3 T-type

(3 × 1) (3)

[16]

QUESTION 3: TRANSMISSION LINES

3.1 Explain how the reactive component can be tuned out of a transmission line and use a simple sketch to support your description. (4)

3.2 Give a brief description of how a Smith chart can be used to obtain the SWR on a transmission line. (5)

3.3 The propagation constant of a transmission line is $\gamma = \alpha + j\beta$

3.3.1 Name the quantity that is represented by α and the unit in which it is measured.

3.3.2 Name the quantity that is represented by β and the unit in which it is measured.

(2 × 2) (4)

3.4 Determine the open-circuit impedance of a transmission line for the following data:

Z_O = 600 ohms
$Z_{S/C}$ = 750 ohms (3)

[16]

QUESTION 4: NOISE

4.1 An FM-receiver has an input resistance of 5 kilo-ohms and an equivalent resistance of 7,2 kilo-ohms. The amplifier has a gain of 20 which is developed across a load resistor of 18 kilo-ohms.

Take Boltzmann's constant as $1,38 \times 10^{-23}$J/K.

Calculate the following:

4.1.1 RMS noise voltage generated at 25 °C and a bandwidth of 20 kHz

4.1.2 Noise figure if the receiver is connected to a signal generator which has an impedance of 300 ohms

(2 × 3) (6)

4.2 Draw the equivalent input circuit for noise of the amplifier in FIGURE 2.

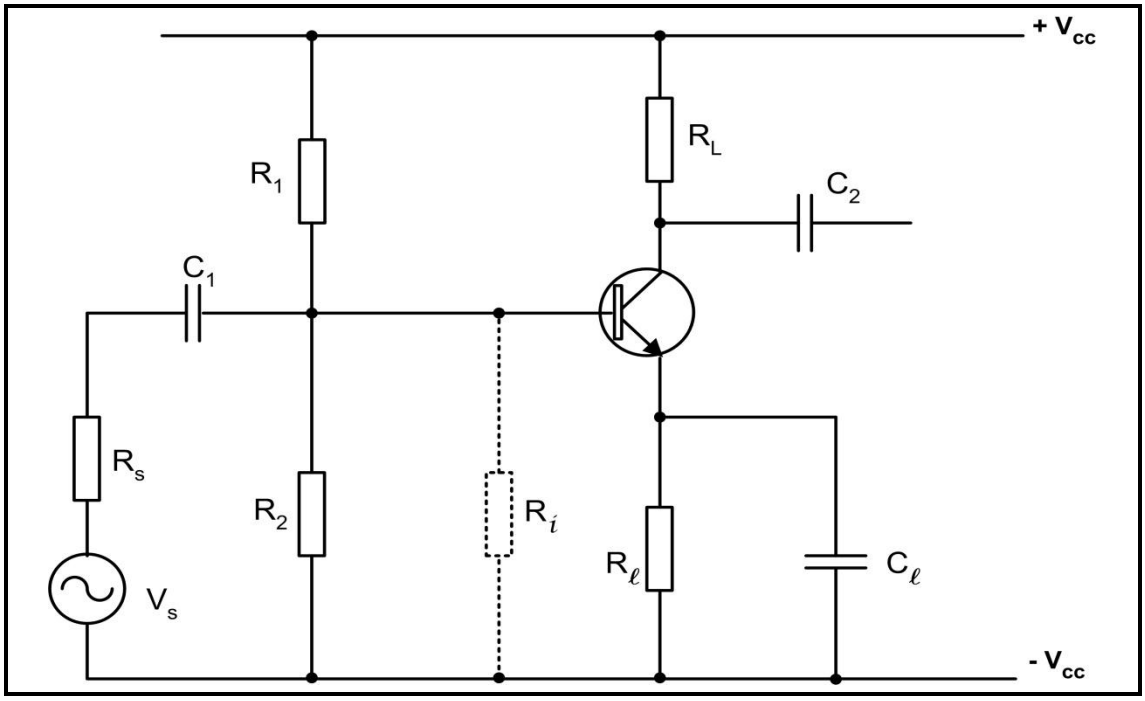


FIGURE 2 (4)

4.3 A radar installation is used to detect a 9 m^2 target at a distance of 100 km. The frequency of operation is 3 GHz and both the transmitter and receiver antennae have a gain of 20 each.

In order to detect the target successfully a minimum power of $1,8 \times 10^{-18}$ watt is needed. There is however an attenuation of 0,002 dB/km.

Determine the minimum transmitter power required to detect this target. (7)

[17]

QUESTION 5: MODULATION AND DEMODULATION

5.1 Draw a neat, labelled circuit diagram of a class C modulator using an NPN transistor. Indicate ALL the relevant waveforms. (7)

5.2 Discuss the full operation of the class C modulator in QUESTION 5.1. (5)

5.3 Define each of the following with reference to modulation:

5.3.1 PAM

5.3.2 PPM

5.3.3 PWM

(3 × 2) (6)
[18]

QUESTION 6: ANTENNAS

6.1 Define *bandwidth* of an antenna. (2)

6.2 A satellite transmitter operating at a frequency of 950 MHz radiates 60 kW of RF power and is used to determine the distance of a satellite 150 km away with a surface area of 30 m^2. The transmitter and receiver antennae have similar characteristics with gains of 15 dB each.

Determine each of the following:

6.2.1 Field strength at the satellite (3)

6.2.2 Power density at the satellite (2)

6.2.3 Reflected power from the satellite (2)

6.2.4 Field strength at the receiver (2)

6.2.5 Power density at the receiver (2)

6.2.6 Received power at the receiver (3)
[16]

TOTAL: 100

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COMMUNICATION-ELECTRONICS N6

FORMULA SHEET

1. Filters

Low-pass/Laedeurlaat

$$f_c = \frac{1}{\pi \sqrt{L \,.\, C}} \qquad Z_o = \sqrt{\frac{L}{C}}$$

High-pass/Hoëdeurlaat

$$f_c = \frac{1}{4 \,.\, \pi \sqrt{L \,.\, C}} \qquad Z_o = \sqrt{\frac{L}{C}}$$

Low-pass m-derived sections (also for half-sections)
Laedeurlaat-m-afgeleideseksies (ook vir halfseksies)

T-section *T-seksie*	*π-section* *π-seksie*
Series/Serie $L = \frac{m \,.\, L}{2}$	*Shunt/Sjunt* $C = \frac{m \,.\, C}{2}$
Shunt/Sjunt $C = m \,.\, C$	*Series/Serie* $L = m \,.\, L$
Shunt/Sjunt $L = \frac{1 - m^2}{4 \,.\, m} \,.\, L$	*Series/Serie* $C = \frac{1 - m^2}{4 \,.\, m} \,.\, C$
$m = \sqrt{1 - \frac{f_c^2}{f_\infty^2}}$ $L = \frac{Z_o}{\pi \,.\, f_c}$	$C = \frac{1}{\pi \,.\, Z_o \,.\, f_c}$

High-pass m-derived sections (also for half-sections)
Hoëdeurlaat-m-afgeleideseksies (ook vir halfseksies)

T-section *T-seksie*	*π-section* *π-seksie*
Series/Serie $C = \frac{2 \,.\, C}{m}$	*Series/Serie* $C = \frac{C}{m}$
Shunt/Sjunt $C = \frac{4 \,.\, m}{1 - m^2} \,.\, C$	*Series/Serie* $L = \frac{4 \,.\, m}{1 - m^2} \,.\, L$
Shunt/Sjunt $L = \frac{L}{m}$	*Shunt/Sjunt* $L = \frac{2 \,.\, L}{m}$

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$$m = \sqrt{1 - \frac{f_\infty^2}{f_c^2}} \quad L = \frac{Z_o}{4 \,.\, \pi \,.\, f_c} \qquad C = \frac{1}{4 \,.\, \pi \,.\, Z_o \,.\, f_c}$$

Band-pass sections
Banddeurlaatseksies

$$\textit{Series/Serie}\, L = \frac{Z_o}{2 \,.\, \pi \,[f_2 - f_1]} \qquad \textit{Series/Serie}\, C = \frac{f_2 - f_1}{2 \,.\, \pi \,.\, Z_o \,.\, f_1 \,.\, f_2}$$

$$\textit{Shunt/Sjunt}\, L = \frac{Z_o \,[f_2 - f_1]}{4 \,.\, \pi \,.\, f_1 \,.\, f_2} \qquad \textit{Shunt/Sjunt}\, C = \frac{1}{\pi \,.\, Z_o \,[f_2 - f_1]}$$

Band-stop sections
Bandsperseksies

$$\textit{Series/Serie}\, L = \frac{Z_o \,[f_2 - f_1]}{2 \,.\, \pi \,.\, f_1 \,.\, f_2} \qquad \textit{Series/Serie}\, C = \frac{1}{2 \,.\, \pi \,.\, Z_o \,[f_2 - f_1]}$$

$$\textit{Shunt/Sjunt}\, L = \frac{Z_o}{4 \,.\, \pi \,.\, [f_2 - f_1]} \qquad \textit{Shunt/Sjunt}\, C = \frac{f_2 - f_1}{\pi \,.\, Z_o \,.\, f_1 \,.\, f_2}$$

2. *Attenuators*
Verswakkers

Symmetrical
Simmetries

Bridged T-type
Gebrugde T-tipe

$$Z_B = Z_o \,[N - 1]$$

$$Z_c = \frac{Z_o}{N - 1}$$

$$Z_A = Z_o$$

$$\textit{where/waar}\; dB = 20 \log N$$

Asymmetrical
Asimmetries

L-type
L-tipe

$$Z_A = \sqrt{R_1 \,(R_1 - R_2)}$$

$$Z_B = \sqrt{\frac{(R_1 \,.\, R_2^2)}{R_1 - R_2}}$$

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T-Type
T-Tipe

$$Z_1 = R_1\left[\frac{N^2+1}{N^2-1}\right] - 2\sqrt{R_1 \,.\, R_2}\left[\frac{N}{N^2-1}\right]$$

$$Z_2 = 2\sqrt{R_1 \,.\, R_2}\left[\frac{N}{N^2-1}\right]$$

$$Z_3 = R_2\left[\frac{N^2+1}{N^2-\ 1}\right] - 2\sqrt{R_1 \,.\, R_2}\left[\frac{N}{N^2-1}\right]$$

where/waar $dB = 20 \log N$

π-type
π-tipe

$$Z_A = R_1\left[\frac{N^2-1}{N^2-2\,.\,N\,.\,S+1}\right]$$

$$Z_B = \frac{\sqrt{R_1 \,.\, R_2}}{2}\left[\frac{N^2-1}{N}\right]$$

$$Z_c = R_2\left[\frac{N^2-1}{N^2-2\,.\,N/S+1}\right]$$

where/waar $dB = 20 \log N$ *and/en* $S = \sqrt{\frac{R_1}{R_2}}$

Insertion loss/Tussenskakelverlies $= 20 \log \frac{I_1}{I_2}$

3. Transmission lines
Transmissielyne

$$Z_o = \sqrt{Z_{o/c} \,.\, Z_{s/c}} \qquad Z_o = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$\gamma = \sqrt{[R + j\omega L]\,[G + j\omega C]} \qquad \gamma = \alpha + j\beta$$

$$\lambda = \frac{2\,.\,\pi}{\beta} \quad V = \frac{\omega}{\beta} \qquad t = \frac{d}{v}$$

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$L.G = C.R$

$VSWR = \frac{Z_R}{Z_o}$ *or/of* $\frac{Z_o}{Z_R}$

Also/ook $VSWR = \frac{1+\rho}{1-\rho}$

$\rho = \frac{Z_R - Z_o}{Z_R + Z_o}$

$\rho = \frac{V_{(reflected/\ weerkaatse)}}{V_{(incident/\ insident)}}$

Normalised impedance/Genormaliseerde impedansie

$Z_o^{'} = \frac{R}{Z_o} \pm j\ \frac{X}{Z_o}$

4. Noise/Geruis

$E_n = \sqrt{4KT\Delta fR}$ $\qquad$ $E_n = \sqrt{4KT\Delta f\ [R_1 + R_2]}$

$R_{eq} = R_A + \frac{R_B}{A_1^2} + \frac{R_c}{A_1^2\ .\ A_2^2}$

where/ waar $R_A = R_{1in} + R_{1eq}$

$R_B = \frac{R_{1(out/uit)}\ .\ R_{2(in)}}{R_{1(out/uit)} + R_{2(in)}} + R_{2(eq)}$

$R_c = R_{2out/uit}$

$F = 1 + \frac{R_{eq}^{'}}{R_a}$ $\qquad$ $R_{eq}^{'} = R_{eq} - R_t$

$F = \frac{S/N_{(in)}}{S/N_{(out/uit)}}$ $\qquad$ $F = 1 + \frac{T_{eq}}{T_o}$

$C = \Delta f \log_2 [1 + S/N]$

where/waar $dB = 10 \log S/N$

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5. Modulation
Modulasie

$$m = \frac{V_m}{V_c} \qquad \eta = \frac{P_{dc}}{P_c} \times 100\%$$

$$P_{dc} = I_{mean} \times V_{dc} \qquad P_{\text{mod}} = \frac{P_{SB}}{\eta}$$

$$P_{SB} = \frac{m^2 \,.\, P_c}{2}$$

$$P_T = P_c + P_{SB}$$

6. Antennae/Antennas

$$P_r = P_d \times A \quad A = \frac{\lambda^2}{4 \,.\, \pi} \qquad \lambda = \frac{c}{f}$$

$$P_r = P_d \times A \,.\, G_r \quad \textit{where/waar} \qquad dB = 10 \log G$$

$$P_d = \frac{E^2}{120 \,.\, \pi} \qquad P_d = \frac{P_t \,.\, G_t}{4 \,.\, \pi \,.\, d^2}$$

$$P_d = \frac{30 \,.\, G \,.\, P_t}{120 \,.\, \pi \,.\, d^2} \qquad P_t = \frac{[4 \,.\, \pi]^3 \,.\, d^4 \,.\, P_r}{G_t \,.\, G_r \,.\, Q \,.\, \lambda^2}$$

$$P_r = \frac{10 \,.\, P_t \,.\, G_t}{4 \,.\, \pi \,.\, d^2} \qquad E_r = \frac{\sqrt{30 \,.\, P_r}}{d}$$

$$E_S = \frac{\sqrt{30 \,.\, P_t \,.\, G_t}}{d} \qquad A = \frac{G \,.\, \lambda^2}{4 \,.\, \pi}$$

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